

Python OOP

Practice Questions — Easy to Hard

30 Questions across 6 Topics

1■■■ Class, Object & Constructor

● Easy

Q1. Dog class banao — name, breed, age. Method bark() → 'Woof! Mera naam Bruno hai'. Object banao aur bulao.

● Easy

Q2. Rectangle class — parameterized constructor length, width. Methods: area(), perimeter(). 3 objects banao alag dimensions se.

● Medium

Q3. Student class — name, marks (list of 5 subjects). Constructor me calculate karo total aur percentage. Method result() → pass if percentage ≥ 40 else fail.

● Medium

Q4. Counter class — default value constructor start=0, step=1. Methods: increment(), decrement(), reset(), show(). 2 objects alag step se test karo.

● Hard

Q5. Hospital class — name, doctors (list), patients (list). Methods: add_doctor(), add_patient(), remove_patient(), show_all(). Constructor me validate karo — hospital name empty nahi hona chahiye.

2■■■ Inheritance

● Easy

Q1. Animal → Dog single inheritance. Animal me eat(), sleep(). Dog me bark(). Dog object se teeno bulao.

● Easy-Medium

Q2. Person → Student → Topper multilevel. Har level me ek method. super() use karke upar ka bhi print karo.

● Medium

Q3. Flyable, Swimmable, Runnable — teen alag classes. Triathlete class teeno se inherit kare. Har method bulao.

● Medium

Q4. Vehicle → Car, Truck, Bike hierarchical. Vehicle me fuel_type, speed. Har child me extra attribute aur describe() method — super() use karo.

● Hard

Q5. SmartPhone(Phone, Camera, MusicPlayer) — multiple + hybrid. Har parent me 2 methods. SmartPhone me super() se sab bulao + apna assistant() method.

3■■ Polymorphism

● Easy

Q1. Shape → Circle, Square, Triangle. Har child me draw() override kare. List me teeno rakho, loop se draw() bulao.

● Easy-Medium

Q2. Calculator class — operate(*args): 1 arg → square, 2 args → add, 3 args → multiply teeno.

● Medium

Q3. Notification — send(message) abstract. EmailNotification, SMSNotification, PushNotification implement kare. notify_all(notifications, msg) function — sab ko bhejo.

● Medium

Q4. Book class — __gt__, __lt__, __eq__ pages se compare karo. 5 books ki list banao, max() aur min() se sabse moti/patli book nikalo.

● Hard

Q5. FileConverter duck typing — PDFConverter, WordConverter, ImageConverter. Har ek me read(), convert(), save(). batch_convert(converters) function — list me sab converters lo, ek ek process karo.

4■■ Encapsulation

● Easy

Q1. Person class — name public, _phone protected, __aadhar private. Getter + setter __aadhar ke liye — setter me validate karo (12 digits hona chahiye).

● Easy-Medium

Q2. Locker class — __pin private. Methods: set_pin(pin), verify_pin(pin), open_locker(pin) → sahi pin tabhi open.

● Medium

Q3. Student class — __marks private dict ({subject: marks}). @property: marks getter, marks setter (0-100 validate), average property, highest property → max marks wala subject.

● Medium

Q4. Inventory class — __stock private dict. Methods: add_item(name, qty), remove_item(name, qty) → stock 0 se kam nahi hona chahiye, @property stock → return karo.

● Hard

Q5. SecureUser class — `__password` private, `__login_attempts` private (max 3). Methods: `set_password()`, `login(password)` → 3 baar galat → account lock, `reset_attempts()`, `is_locked` property.

5■■■ Abstraction

● Easy

Q1. Appliance abstract — `turn_on()`, `turn_off()`. Fan, AC, WashingMachine implement kare. Sab ka on/off test karo.

● Easy-Medium

Q2. Logger abstract — `log(message)`. FileLogger, ConsoleLogger, DatabaseLogger implement kare. `log_all(loggers, msg)` function banao.

● Medium

Q3. CloudStorage abstract — `upload(file)`, `download(file)`, `delete(file)`. GoogleDrive, Dropbox implement kare. `sync(storage, files)` function — list of files upload karo.

● Medium

Q4. Game abstract — `start()`, `play()`, `end()`. Chess, Cricket, Ludo implement kare. `run_game(game)` function → start → play → end sequence.

● Hard

Q5. DataProcessor abstract — `load()`, `process()`, `save()`, `generate_report()`. CSVProcessor, JSONProcessor implement kare. Constructor me filename lo. `pipeline(processor)` function — sab steps chalao aur time print karo (import time).

6■■■ Dunder Methods

● Easy

Q1. Phone class — `brand`, `storage`. `__str__`, `__repr__`, `__eq__` (same storage = equal), `__gt__` (zyada storage = bada).

● Easy-Medium

Q2. Wallet class — `balance`. `__add__` (wallets merge), `__sub__` (balance ghataao), `__str__`, `__bool__` → `balance > 0` tabhi True.

● Medium

Q3. Playlist class — `songs` list. `__len__`, `__getitem__`, `__contains__`, `__str__`, `add(song)`, `remove(song)`.

● Medium

Q4. Fraction class — `numerator`, `denominator`. `__add__`, `__sub__`, `__mul__`, `__str__` → '3/4', `__eq__`. Simplify karo gcd se.

● Hard

Q5. DataFrame class — 2D dict {column: [values]}. `__len__` → rows, `__getitem__` → column data, `__contains__` → column exist?, `__str__` → table format print, `add_column(name, data)`, `add_row(data)`.

Ek ek karke solve karo — Claude check karega! ■